CSC 3730 — Artificial Intelligence for Simulations

Overview

Course Information

Course	CSC 3730 -- Artificial Intelligence for Simulations
Semester	AY 2026-2027 Fall
Credits	4
Days / Time	TR 9:50-11:30am
Location	DSC 99
Prerequisites	CSC 1810 or equivalent

Instructor Information

Instructor	Dr. Jacob Pichelmeyer PhD
Email	jpichelmeyer @carthage.edu
Office	DSC 72
Office Hours	M 10:30am- 1:30pm T 11:35am-12:35am W 10:30am-11:30am

Course Description

Explore the fundamental AI algorithms used in simulations and game development. This course covers techniques like pathfinding, decision trees, behavior trees, finite state machines, and machine learning. Students will apply these algorithms to create more dynamic, responsive, and intelligent virtual environments. Ideal for those interested in game design, simulations, and AI programming.

Student Learning Outcomes

1. Apply AI algorithms such as pathfinding, decision trees, and behavior trees to create dynamic and responsive simulations.
2. Analyze the performance and effectiveness of different AI techniques in game and simulation environments.
3. Design intelligent behaviors for virtual environments by implementing AI-driven systems and solutions.
4. Evaluate the use of machine learning algorithms in simulations, determining their suitability for specific scenarios.
5. Create a fully functional AI-drven simulation project that demonstrates the integration of multiple AI techniques.

Instructional Methods

Lecture, live-coding demo, hands-on studio/lab work.

Course Materials

Laptop, writing utensils, paper

Course Assignment, Assessments, and Grading Policy

Grading Policy

{'A': '93', 'A-': '90', 'B+': '87', 'B': '83', 'B-': '80', 'C+': '77', 'C': '73', 'C-': '70', 'D+': '67', 'D': '63', 'D-': '60', 'F': '0'}

Assessments

Assessment	Weight	Description
Micro-quiz	5%	Given spontaneously to assess understanding of something very recently discussed. Short (5 min or less). Small stakes.
Presentation 1: The Pitch	10%	Students pitch their simulation world: what living system or phenomenon they want to model, why it interests them, and what their agent will do in it.
Presentation 2: The Progress	20%	Students present the current state of their semester-long project.
Presentation 3: The...Perfection?	30%	Students present the 'final' state of their semester-long project.
Progress Reports	20%	At the end of each week, students will turn in a progress report, detailing what they have accomplished on their term project since the prior report. Must be *hand-written.*
Quizzes	15%	A short in-class quiz each week on the prior week's reading and concepts.
Total	**100%**	

Grading Scale

Letter Grade	Percentage Range
A	93% - 100%
A-	90% - 92%
B+	87% - 89%
B	83% - 86%
B-	80% - 82%
C+	77% - 79%
C	73% - 76%
C-	70% - 72%
D+	67% - 69%

D	63% - 66%
D-	60% - 62%
F	Below 60%

Course Policies and Procedures

Attendance Policy

Students are expected to attend class. Though no formal attendance will be taken, many days will feature a microquiz, which will act as a form of attendance for that day.

Expectations for Classroom Behavior

We all thrive when everyone feels safe, respected, and included. Bigotry, bullying, or mistreatment of any kind will not be tolerated in our classroom.

Late Work/Make-up Policy

Late work will not be accepted.

Academic Integrity Statement

It is expected that students will abide by the college's Academic Honesty Policy.

Artificial Intelligence Use Policy

AI, like any technology, carries risks and benefits. It is to your advantage to understand both clearly.

Risks: heavy, *unreflective* AI use is linked to *cognitive offloading.* Research suggests that relying on AI to produce answers, rather than to support your own reasoning, can reduce engagement with the thinking process itself and may weaken independent problem-solving over time. [1] [2]

Benefits: When used as a Socratic partner, debugger, or explainer rather than an answer generator, AI has been shown to improve critical thinking, argumentation, and conceptual understanding. Students who actively critique AI output rather than accepting it at face value tend to learn more than students who simply defer to it. [1] [2]

Learning to use AI well is a valuable skill, but it cannot replace your independent reasoning, judgment, and creativity. While companies use AI to automate routine knowledge work, human value lies in the ability to critique these tools, know when to question them, and go beyond what they can produce. This class is your opportunity to develop those unique, non-automated strengths. Ultimately, your worth is not defined by economic productivity or outperforming a machine. Rather, the goal for this class is to cultivate a deep conceptual understanding and human agency that algorithms will struggle to replicate.

All work you turn in is expected to be your own. In order to earn full points on assessments, you must be able to fully and clearly explain all aspects of your work.

Student Support Resources

Student-Athletes

If you are a student athlete, you should inform me of this status at the start of the semester by providing me with the Coach's Letter that you received (this letter should name your head coach and the competition dates, and should have been provided to you at the first meeting). In addition to sharing this information at the start of the semester, you need to notify me of any class days to be missed due to a Carthage Athletics-sponsored event in which you are participating. You should notify me in advance of each upcoming contest. (For example, if you know you are traveling with the team on Friday, please let me know several days in advance). You will need to make up any missed lectures, assignments, and/or exams. Some work that can happen only during class time, such as discussion points, is not possible to make up. Students are encouraged to be in communication often; excessive absences will necessitate conversations with the instructor and the associate athletic director, and may impact your grade. Please note: student-athletes cannot miss class to attend a practice session or athletic training appointment.

Learning Accessibility Service Information

Carthage College strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers due to your particular circumstances (including mental health, learning disorders, and chronic medical conditions), please let me know immediately so we can discuss options privately. To establish reasonable accommodations, you must register with Warren Wolchuk in Learning Accessibility Services (wwolchuk@carthage.edu).

The Health and Counseling Center

The Health and Counseling Center (HCC) addresses student physical, mental, and emotional well-being. All services, provided by experienced professionals, are free and confidential to currently enrolled, full-time undergraduate students. Students must call or visit the HCC to schedule an appointment. Health services are available by walk-in or appointment from 8:30am to 1pm and 2pm to 4:00pm. Counseling walk-in sessions are available Monday through Friday from 11:30am to 1:00pm. Last walk-in appointment begins 30 minutes before the end of walk-in hours. TWC, first floor (behind mailboxes), 262-551-5710. Students can visit Uwill to access teletherapy services.

Writing Center

The Brainard Writing Center is a free resource for student writers. The center is staffed by undergraduate Writing Fellows who have been recommended by Carthage faculty and trained to work with other students on their writing. They can work with you at all stages of the writing process, including understanding the assignment, brainstorming ideas, drafting, revising, and proofreading. Appointments are available in-person, via Zoom, or as written feedback, and can be scheduled on WCOnline. For more information, visit carthage.edu/writing-center.

Peer Tutoring

Peer Tutoring is available to assist you with any aspect of the class, including understanding readings, preparing for quizzes, and studying for exams. This free campus resource is here to help everyone maximize their academic potential. Please take advantage of it. You can schedule an appointment up to seven days in advance and see the schedule on WCOnline. For more information, visit carthage.edu/tutoring.

Course Specifics

Workload Expectation

For every credit this course is worth, it is expected that you will spend 2-3 hours *outside of our in-class time* on work related to the course, which includes (but is not limited to) studying lessons, working on projects, and seeking assistance from and working with the instructor during office hours. This course is worth **4 credit(s)**, so plan on roughly **44 to 444 hour(s)** of outside effort per week. This number may be much higher or lower, depending on the knowledge and experience you bring into the course.

Calendar of Course Activities

Week	Topic
Movement	
Week 1	Situated Agent Model (SAM), Setup, Godot, GDscript
Week 2	Steering behavior, Obstacles, Flocking (boids)
Week 3	Grid & Graph, Pathfinding, Search on graphs
Week 4	Ant Colony Optimization (ACO), Artificial Life
Week 5	Presentation 1: The Pitch
Decision-making	
Week 6	Finite state machines
Week 7	Decision trees, Behavior trees
Week 8	Behavior tree arboriculture
Week 9	Behavior trees vs FSMs
Week 10	Presentation 2: The Progress
Learning	
Week 11	Machine learning overview, Classification

Week 12	Artificial neural networks
Week 13	Reinforcement learning
Week 14	BREAK
Week 15	Project assist
Week 16	Presentation 3: The...Perfection?